A guide for calculating study-level statistical power for meta-analyses

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Abstract

Meta-analysis is a popular approach in the psychological sciences for synthesizing data across studies. However, the credibility of meta-analysis outcomes depends on the evidential value of studies included in the body of evidence used for data synthesis. One important consideration for determining a study's evidential value is the statistical power of the study's design and statistical test combination for detecting hypothetical effect sizes of interest. Studies with a design/test combination that cannot reliably detect a wide range of effect sizes are more susceptible to questionable research practices and exaggerated effect sizes. Therefore, determining the statistical power for design/test combinations for studies included in meta-analyses can help researchers make decisions regarding confidence in the body of evidence. As the one true population effect size is unknown when hypothesis testing, an alternative approach is to determine statistical power for a range of hypothetical effect sizes. This tutorial introduces the *metameta* R package and web app, which facilitates the straightforward calculation and visualization of study-level statistical power in meta-analyses for a range of hypothetical effect sizes. Readers will be shown how to re-analyze data using information typically presented in meta-analysis forest plots or tables and how to integrate the *metameta* package when reporting novel meta-analyses. A step-by-step companion screencast video tutorial is also provided to assist readers using the R package.

Statistical power is the probability that a study design and statistical test combination can detect hypothetical effect sizes of interest. An *a priori* power analysis is often used to determine a sample size (or observation number) parameter using three other parameters: a desired power level, hypothetical effect size, and alpha level. As any one of these four parameters are a function of the remaining three parameters, statistical power can also be calculated using the parameters of sample size, alpha level, and hypothetical effect size. It follows that when holding alpha level and sample size constant, statistical power decreases as the hypothetical effect size decreases. Therefore, one can compute the *range* of effect sizes that can be reliably detected (i.e., those associated with high statistical power) with a given sample size and alpha level. For instance, a study design with sample size of 40 and an alpha of .05 (two-tailed) that uses a paired samples *t*-test has an 80% chance to

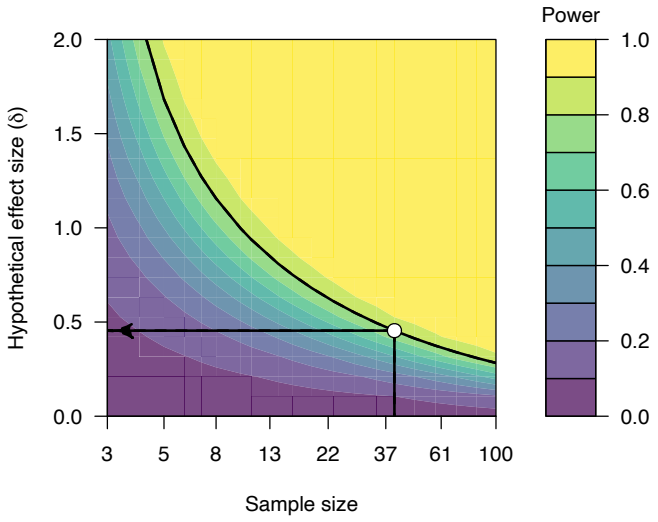


Fig. 1. When holding sample size and alpha level constant, the chances of reliably detecting an effect (i.e., power) depends on the hypothetical true effect size. For a within-participants study with 40 participants that uses a paired samples *t*-test to make inferences, there is an 80% chance of detecting an effect size of 0.45. These chances decrease with smaller hypothetical effect sizes, Figure created using the 'jpower' JAMOVI module (https://github.com/richarddmorey/jpower).

detect an effect size of 0.45, but only have 50% chance of detecting of an effect size of 0.32 (**Fig. 1**). In other words, this study design and test combination would have a good chance of missing effect sizes smaller than 0.45.

In addition to having a lower probability of discovering true effects (Button et al., 2013), study design/test combinations that cannot reliably detect a wide range of effects also have a lower probability that statistically significant results represent true effects (Ioannidis, 2005). In addition, such study design/test combinations tend to be associated with questionable research practices (Dwan et al., 2008), and are more likely to report exaggerated effect sizes (Ioannidis, 2008; Rochefort-Maranda, 2021). In light of these factors, the contribution of low statistical power to the reproducibility crisis in the psychological sciences has become increasingly recognized (Button et al., 2013; Munafò et al., 2017; Walum et al., 2016). However, despite meta-analysis being often considered the gold-standard of evidence (but see Stegenga, 2011), the role of study-level statistical power for calculating effect sizes in meta-analysis outcomes is rarely considered. This can be a critical oversight as studies included in a meta-analysis that are not designed to reliably detect meaningful effect sizes have reduced evidential value, which diminishes confidence in the body of evidence. While inverse variance weighting and related approaches can reduce the influence of studies with larger variance (i.e., those with less statistical power) on the summary effect size estimate, these procedures only attenuate the influence of studies that have larger variances *relative* to other studies in the meta-analysis. Moreover,

this attenuation can be quite modest for random effects meta-analysis, which is the dominant meta-analysis model in the psychological sciences. Evaluating statistical power for study heterogeneity and moderator tests (Hedges & Pigott, 2004; Huedo-Medina et al., 2006) can also be used to help determine the *overall* evidential value of a meta-analysis (Bryan et al., 2021; Linden & Hönekopp, 2021), however, these analyses are beyond the scope of this article and associated R package.

One possible reason for the lack of consideration of study-level statistical power in meta-analysis is that it can be time consuming to calculate statistical power for a body of studies if data were to be directly extracted from each study. A recently proposed solution for calculating study-level statistical power is the sunset (power-enhanced) plot (**Fig. 2**), which is a feature of the *metaviz* R package (Kossmeier et al., 2020). While sunset plots are informative as they visualize the statistical power for all studies included in a meta-analysis, they can only visualize statistical power for one effect size of interest at a time. By default, this effect size is the observed summary effect size calculated for the associated meta-analysis (although statistical power for any single effect size of interest can be calculated).

Despite the utility of sunset plots there are some limitations associated with a single effect size approach. First, unless the meta-analysis is only comprised of Registered Report studies (Chambers & Tzavella, 2021) it is very likely that the observed summary effect size is inflated due to publication bias (Ioannidis, 2008;

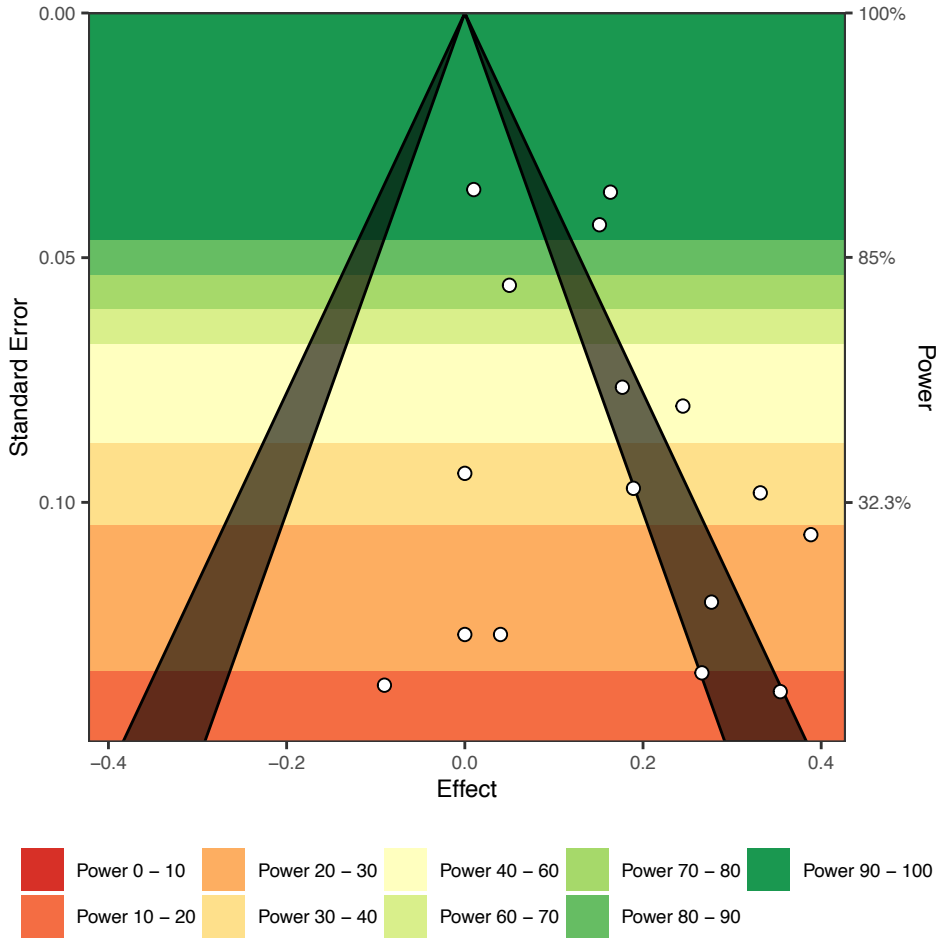


Fig. 2. Sunset plots visualize the statistical power for each study included in a meta-analysis for a given hypothetical effect size. The default hypothetical effect size is the observed summary effect size from the associated meta-analysis, but this can be changed to any hypothetical effect size.

Kvarven et al., 2020; Lakens, 2022; Schäfer & Schwarz, 2019). Using Jacob Cohen's suggested threshold levels for a small/medium/large effect as an alternative should be avoided as a first option if possible as these thresholds were only suggested as fallback for when the effect size distribution is unknown (Cohen, 1988). What actually constitutes a small/medium/large effect differs according to subfield (e.g., Gignac & Szodorai, 2016; Quintana, 2016), study population/context (Kraft, 2020), and is also likely to be influenced by publication bias (Nordahl-Hansen et al., 2022). However, publication bias and issues regarding the inaccuracy of effect size

thresholds are essentially moot points as the true effect size is unknown when testing hypotheses (Lakens, 2022). Alternatively, researchers can determine the *range* of effect sizes a study design can reliably detect and evaluate whether this range includes meaningful effect sizes. In most cases, determining what constitutes a meaningful effect is not a straightforward task, as typical effect sizes vary from field-to-field and researchers can understandably draw different conclusions. Presenting statistical power assuming a range of true effect sizes, instead of a single true effect size, allows readers to transparently evaluate study-level statistical power according to their own assumptions and what is reasonable for a given research field.

The *metameta* package has been developed to address these limitations by calculating and visualizing study-level statistical power for a range of hypothetical effect sizes. Along with calculating statistical power for a range of hypothetical effect sizes for each individual study, a median is also calculated across studies to provide an indication of the evidential value of the body of evidence used in a meta-analysis. There are two broad use cases for the *metameta* package. The first is the re-evaluation of published meta-analysis (e.g., Cherubini & MacDonald, 2021; Quintana, 2020). This could either be for individual meta-analyses or for pooling several meta-analyses on the same topic or in the same research field. Pooling meta-analysis data into a larger analysis is also known as a meta-meta-analysis (hence the package name *metameta*, as this was the original motive for developing the package). The second use case is the implementation of the *metameta* package when reporting

novel meta-analyses (Boen et al., 2022; Gallyer et al., 2021). For example, Gallyer and colleagues (2021) performed a meta-analysis on the link between event related potentials derived from electroencephalograms and suicidal thoughts and behaviors, finding no significant relationships. An early implementation of *metameta* package was used to complement this analysis, which revealed that most included studies were considerably underpowered to detect meaningful effect sizes for the field. Considering this result, the authors concluded that the quality of the evidence was not sufficient to confidently determine the absence of evidence for a relationship. The *metameta* package is also relevant for helping address checklist item 15 in the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 checklist—methods used to assess confidence in the body of evidence (Page et al., 2021)—when reporting meta-analyses.

This purpose of this article is to provide a non-technical introduction to the *metameta* package. The R script used in this article and example datasets can be found on this article's Open Science Framework (OSF) Page https://osf.io/dr64q/. For readers that are not familiar with R, a companion web app is available at https://dsquintana.shinyapps.io/metameta_app/. This article's OSF page also contains the R script used to generate the web browser application for download, which can be used to run the application locally without requiring persistent web access. A screencast video with step-by-step instructions for using the *metameta* package is also provided at https://bit.ly/3Rol42f and the article's OSF page.

Package overview

The *metameta* package contains three core functions for calculating and visualizing study-level statistical power in meta-analyses for a range of hypothetical effect sizes (**Fig. 3**). The **`mapower_se()`** and **`mapower_ul()`** functions perform study-level statistical power calculations and the **`firepower()`** function creates a visualization of these results. The **`mapower_se()`** function uses standard error data, whereas the **`mapower_ul()`** function uses 95% confidence interval data to calculate the

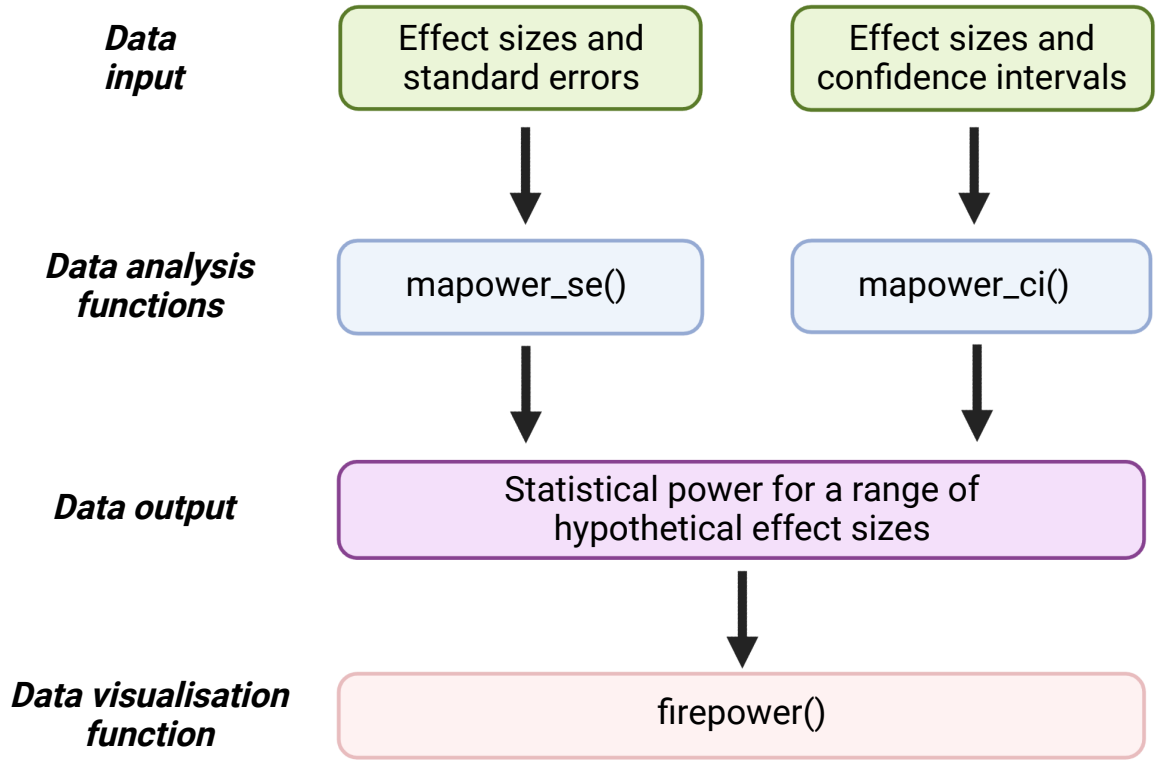


Fig. 3. The ***metameta*** package workflow for calculating and visualizing study-level statistical power for a range of hypothetical effect sizes. Data can be imported either with standard errors or confidence intervals as the measure of variance, which determines whether the **`mapower_se()`** or **`mapower_ci()`** function is used. Both functions will calculate statistical power for a range of hypothetical effect sizes and produce output that can be used for data visualization via the **`firepower()`** function.

statistical power associated with a set of studies. The benefit of using standard error and confidence intervals as measures of variance is that at least one of these measures is almost always included in forest plot visualizations generated by popular meta-analysis software packages. The **`firepower()`** function uses output from both these calculator functions (**Fig. 3**). A **`ci_to_se()`** helper function is also included in *metameta*, which converts 95% confidence intervals to standard errors if the user would prefer to use the **`mapower_se()`** function.

The *metameta* package (requiring R version 3.5 or higher) can be loaded using the following commands, which installs the *devtools* package, downloads the *metameta* package from Github, and loads the package:

```
R>   library(devtools)
     devtools::install_github("dsquintana/metameta")
     library(metameta)
```

These packages rely on other R packages that may or may not be installed on your system. In some cases, you might be asked if you would like to update existing R packages on your system or if you want to install from sources the package which needs compilation. It has been recommended that you select "no" in response to both these prompts (Harrer et al., 2021), unless an update is required for the package to operate. If you are not familiar or comfortable with R, a point-and-click web app version of the package is available

(https://dsquintana.shinyapps.io/metameta_app/), which is covered in more detail below.

Three meta-analysis datafiles are also included for demonstration purposes. These meta-analyses synthesize data evaluating the effect of intranasal oxytocin administration on various behavioral and cognitive outcomes, with positive values indicative of intranasal oxytocin having beneficial effects on outcome measures. Oxytocin is a hormone and neuromodulator produced in the brain, which been the subject of considerable research in the psychological sciences interest due to its therapeutic potential for addressing social impairments (Jurek & Neumann, 2018; Leng & Leng, 2021; Quintana & Guastella, 2020). However, this field of research has been associated with mixed results (Alvares et al., 2017), which has partly been attributed to study designs with low statistical power (Quintana, 2020; Walum et al., 2016). The dataset object **`dat_bakermans_kranenburg`** contains effect size and standard error data from a meta-analysis of 19 studies investigating the impact of intranasal oxytocin administration on clinical-related outcomes in samples diagnosed with various psychiatric illnesses (Bakermans-Kranenburg & Van Ijzendoorn, 2013). The dataset object **`dat_keech`** includes effect size and confidence interval data from a meta-analysis on 12 studies investigating the impact of intranasal oxytocin administration on emotion recognition in neurodevelopmental disorders (Keech et al., 2018). Finally, the dataset object **`dat_ooi`** includes effect size and standard error data extracted from a meta-analysis of 9 studies investigating the impact of

intranasal oxytocin administration on social cognition in autism spectrum disorders (Ooi et al., 2016). These three datasets are also available on the article's OSF page https://osf.io/dr64q/.

Calculating study-level statistical power for published studies

When normally distributed effect sizes (e.g., Hedges *g*, Cohen's *d*, Fisher's Z, log risk-ratio) and their standard errors are available, the statistical power of their study designs for a hypothetical effect size can be calculated using a two-sided Wald test. Some commonly used effect sizes that are *not* normally distributed include Pearson's correlation coefficients, risk ratios, and odds ratios. Although the transformation of these effect size metrics into normally distributed effect sizes is relatively straightforward and typical practice for meta-analysis (Borenstein et al., 2021; Harrer et al., 2021), these untransformed effect size metrics are sometimes presented in tables or forest plots even if transformed effect sizes are used for meta-analytic synthesis. Thus, *metameta* users should be wary of this and transform these effect sizes if necessary (Harrer et al., 2021).

The use of the **`mapower_se()`** function for calculating study-level statistical power for a range of effect sizes will be illustrated first. The **`mapower_se()`** function requires the user to specify a minimum of three arguments: `mapower_se(dat, observed_es, name)`. The first argument (dat) is the

dataset that contains one column named 'yi' (effect size data), and one column named 'sei' (standard error data). The second argument (observed_es) is the observed summary effect size of the meta-analysis. If effect size standard errors are not reported, under some circumstances these can be calculated if sample size information is provided (see Appendix A for guide on calculating standard errors for both Cohen's *d* for between-group designs and Pearson's *r* when sample size information is available). While *metameta* calculates statistical power for a range of hypothetical effect sizes, the statistical power of the observed summary effect size is often of interest for comparison to the full range of effect sizes, so this is presented alongside the statistical power for a range of effect sizes when using the **`firepower()`** function, which will be described soon. The third argument (name) is the name of the meta-analysis (e.g., the first author of the meta-analysis), which is used for creating labels when visualizing the data when applying the **`firepower()`** function. Data from the **`dat_ooi`** dataset object will be used (i.e., Hedges' *g*, and standard error), which was extracted from figure 2 from Ooi and colleagues' article (Ooi et al., 2016). Assuming the *metameta* package is loaded using the command described above (also see the analysis script: https://osf.io/dr64q/), the following R script will calculate study-level statistical power for a range of effect sizes and store this in an object called 'power_ooi':

```
R>   power_ooi <- mapower_se(
```

```
dat = dat_ooi,
observed_es = 0.178,
name = "Ooi et al 2017")
```

Note that the observed effect size (observed_es) of 0.178 was extracted from forest plot in figure 2 of Ooi and colleagues' article (2016).

The object 'power_ooi' contains two dataframes. The first dataframe, which can be recalled using the `power_ooi$dat` command, includes the inputted data, statistical power assuming that the observed summary effect size is the true effect sizes, and statistical power for a range of hypothetical effect sizes, ranging from 0.1 to 1. This range is selected as the default as the majority of reported effect sizes in psychological sciences (Szucs & Ioannidis, 2017) are between 0 and 1, although this range can be adjusted (see below). This information is presented in **Table 1,** with the last six columns removed here for the sake of space. These results suggest that none of the included studies could reliably detect effect sizes even as large as 0.4, as with the highest statistical power of 44%. In other words, the study design with highest statistical power (i.e., study 9) would only have a 44% probability of detecting an effect size of 0.4 (assuming an alpha of 0.05 and a two-tailed test).

Table 1. Study level statistical power for a range of effect sizes and the observed effect size for the meta-analysis reported by Ooi and collegues

Study number	study	yi	sei	power_es_observed	power_es01	power_es02	power_es03	power_es04
1	anagnostou_2012	1.19	0.479	0.066	0.055	0.07	0.096	0.133
2	andari_2010	0.155	0.38	0.075	0.058	0.082	0.124	0.183
3	dadds_2014	-0.23	0.319	0.086	0.061	0.096	0.156	0.241
4	domes_2013	-0.185	0.368	0.077	0.059	0.084	0.129	0.192
5	domes_2014	0.824	0.383	0.075	0.058	0.082	0.123	0.181
6	gordon_2013	-0.182	0.336	0.083	0.06	0.091	0.145	0.222
7	guastella_2010	0.235	0.346	0.081	0.06	0.089	0.14	0.212
8	guastella_2015b	0.069	0.279	0.098	0.065	0.111	0.189	0.3
9	watanabe_2014	0.245	0.222	0.126	0.074	0.147	0.272	0.437

Note: Only effect sizes from 0.1 to 0.4 are shown here to preserve space. yi = effect size; sei = standard error; power_es_observed = statistical power assuming that the observed summary effect size is the "true" effect size; power_es01 = statistical power assuming that 0.1 is the "true" effect size.

The second dataframe, which can be recalled using the `power_ooi$power_median_dat` command, includes the median statistical power across all included studies, for the observed summary effect size and a range of effect sizes between 0.1 and 1. This output reveals that the median statistical power for all studies assuming a true effect size of 0.4 is 21%. Finally, the **`firepower()`** function can be used to create a Firepower plot, which visualizes the median statistical power for a range of effect sizes across all studies included in the meta-analysis. The following command will generate a Firepower plot (**Fig. 4**) for the Ooi and colleagues' meta-analysis: `firepower(list(power_ooi$power_median_dat))`. By default, the **`firepower()`** function generates a figure with a generic "Effect size" label on the x-axis. However, it is possible to create a custom label using the

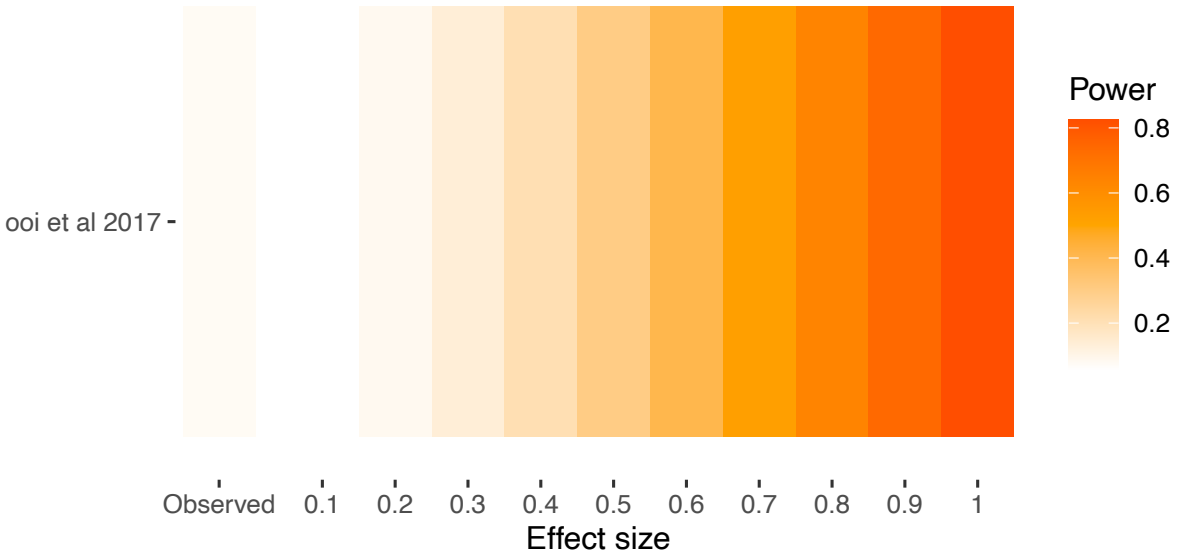


Fig. 4. A Firepower plot, which visualizes the median statistical power for a range of hypothetical effect sizes across all studies included in a meta-analysis. The statistical power for the observed summary effect size of the meta-analysis is also shown.

(es) argument. For example, here is the same script as above that now includes a "Hedges's g" label (figure not shown):

```
R>   firepower(list(power_ooi$power_median_dat),
     es = "Hedges' g").
```

For those who are not familiar with R, the **`mapower_se()`** and **`firepower()`** functions have been implemented in a point-and-click web app https://dsquintana.shinyapps.io/metameta_app/ (**Fig. 5**). To perform the analysis, upload a csv file with effect size and standard error data in the format described above, specify the observed effect size, and name the meta-analysis. From the web app, users can download csv files with analysis results and the Firepower plot as a PDF file.

An in-depth interpretation of these results requires an understanding of what constitutes the smallest effect size of interest (SESOI) for the research question at hand. That is, what is the smallest effect size that is considered worthwhile or meaningful? By determining the SESOI, researchers can determine if a study design/test combination can reliably detect effect sizes that are at least this size, or larger. Of course, resource limitations might play a role (e.g., rare populations), so a researcher might have higher SESOI considering these issues.

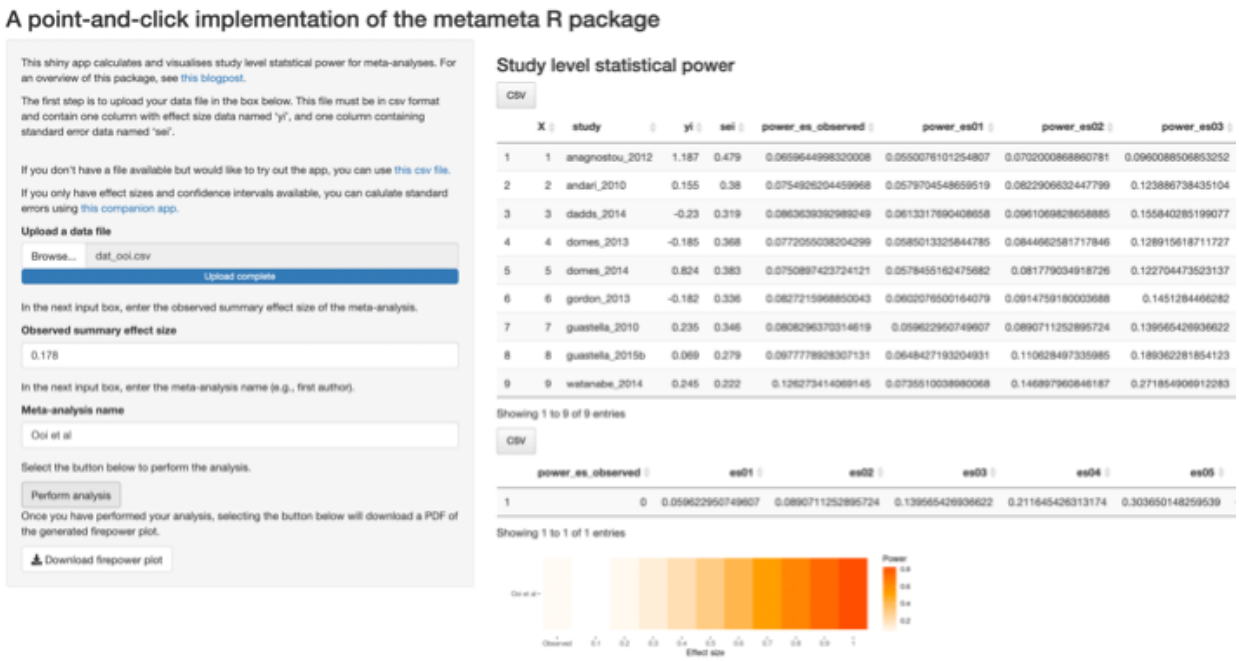


Fig. 5. A screenshot of the metameta web app. Users can upload csv files with effect sizes and standard error data, and the app will calculate study-level statistical power for a range of effect sizes, which can be downloaded as a csv file. A Firepower plot, which visualizes statistical power for a range of effect sizes, will also be generated. The Firepower plot can be downloaded as a PDF file. Note that only the first eight columns for study-level statistical power are shown here for the sake of space.

The use of prior effect sizes reported in the literature is one suggestion among others for determining a SESOI (Keefe et al., 2013; Lakens et al., 2018), which will be used an example for how to interpret information provided by the *metameta* package. For our prior effect size, data from a recent analysis (Quintana, 2020) that indicated that the median effect size across 107 intranasal oxytocin administration trials is 0.14 will be used. This is arguably a conservative estimate as no correction was made for publication bias inflation in this study. With this value of 0.14 in mind, let's return to the results presented in **Table 1**. Even if we were to round up our SESOI from 0.14 to 0.2, none of these included studies would have more than a 15% chance of detecting this effect size. Moreover, there is an increased chance of a

false positive result for the two statistically significant studies included in this meta-analysis (Ooi et al., 2016), which were study 1 and study 9 (**Table 1**). Altogether, if one were to consider 0.14 as the SESOI of interest, both a significant and non-significant meta-analysis result would be largely inconclusive—any non-significant results would have an increased change of being a false negative as these studies were not designed to detect smaller effects, and any significant results are likely to be false positives. For reporting results, one can include the individual study-level data, and group-level data with the associated Firepower plot for visualization (**Fig. 4**). As mentioned above, what constitutes a meaningful or worthwhile effect size (i.e., the SESOI) can differ according to the subfield and a researcher's interpretation. While the *metameta* user can provide their own interpretation of the results based on a justified SESOI, a benefit of the output generated by *metameta* is that it can provide the necessary information for readers to evaluate the credibility of studies or a body of studies according to a SESOI that they have determined.

The default setting for the *metameta* package is for the calculation of power assuming effects that range from 0.1 to 1 in increments of 0.1, which is defined as a "medium" range. While this range is reflects the majority of reported standardized mean differences effect sizes in the psychological sciences (Szucs & Ioannidis, 2017), other ranges might be more appropriate for different subfields, disciplines, or effect size measures (e.g., Pearson's r). Thus, it is possible to specify a smaller or larger range of effect sizes using an additional optional argument. The "small" option

calculates power for a range of effects from 0.05 to 0.5 in increments of 0.05, whereas the "large" option calculates a range of range of effects from 0.25 to 2.5 in increments of 0.25. For example, the following script will perform the same analysis as above, but instead using a smaller range of effect sizes (i.e., 0.05 to 0.5):

```
R>   power_ooi_small <- mapower_se(
     dat = dat_ooi,
     size = "small",
     observed_es = 0.178,
     name = "Ooi et al 2017")
```

By default, the **`firepower()`** function visualizes statistical power for effect sizes ranging from 0.1 to 1, like the **`mapower_se()`** function. However, if the "small" or "large" options are used for the "size" argument, then these effect sizes ranges can also be visualized via the **`firepower()`** function. For example, the 'power_ooi_small' object that we just created above can be used to create a Firepower plot, with a "Hedges' g" label on the x-axis and a "small" effect size range of 0.05 to 0.5 using the following script (figure not shown):

```
R>   firepower(list(power_ooi_small$power_median_dat),
     size = "small",
     es = "Hedges' g").
```

Calculating power with effect sizes and confidence intervals

If a meta-analysis does not report standard error data, it may alternatively present confidence interval data. The **mapower_ul()** function facilitates the analysis of effect size and confidence interval data using the same four arguments as **mapower_se()**, however, the inputted dataset requires a different structure. That is, the **mapower_ul()** function expects a dataset containing one column with observed effect sizes or outcomes labelled "yi", a column labelled "lower" with the lower confidence interval bound, and column labelled "upper" with the upper confidence interval bound. This function assumes a 95% confidence interval was used in the meta-analysis the data was extracted from. To demonstrate the **mapower_ul()** function, data from the **dat_keech** dataset object will be used (i.e., study name, Hedges' *g*, and lower confidence interval, upper confidence interval), which was extracted from figure 2 from Keech and colleagues' article (Keech et al., 2018).

Assuming the *metameta* package is loaded, the following R script will calculate study-level statistical power for a range of effect sizes and store this in an object called 'power_keech':

```
R>   power_keech <- mapower_ul(
          dat = dat_keech,
          observed_es = 0.08,
          name = "Keech et al 2017"
          )
```

The observed effect size (observed_es) of 0.08 was extracted from forest plot in figure 2 of Keech and colleagues' article (Ooi et al., 2016). We can recall a dataframe containing study-level statistical power for a range of effect sizes using the `power_keech$dat` command (**Table 2**), which reveals that at least at the 0.4 effect size level, two studies were designed to reliably detect effects (using the conventional 80% statistical power threshold). However, the median statistical power assuming an effect size of 0.4 (calculated extracted using the `power_keech$power_median_dat` command) was 32%, which is considerably low. As before, we can create a Firepower plot using the following command: `firepower(list(power_keech$power_median_dat)`.

To use the *metameta* web browser application detailed above with confidence interval data, users first need to convert confidence intervals to standard errors using a companion app https://dsquintana.shinyapps.io/ci_to_se/. If both standard error and confidence interval data are available, these will provide equivalent results, perhaps with some very minor differences due to decimal place rounding. However,

Table 2. Study level statistical power for a range of effect sizes and the observed effect size for the meta-analysis reported by Keech and collegues

Study number	study	yi	lower	upper	sei	power_es_o	power_es01	power_es02	power_es03	power_es04
1	anagnostou_2012	0.79	-0.12	1.71	0.467	0.053	0.055	0.071	0.098	0.137
2	brambilla_2016	0.15	-0.22	0.52	0.189	0.071	0.083	0.185	0.356	0.563
3	davis_2013	0.11	-0.68	0.9	0.403	0.055	0.057	0.079	0.115	0.168
4	domes_2013	-0.18	-0.86	0.5	0.347	0.056	0.06	0.089	0.139	0.211
5	einfeld_2014	0.22	-0.06	0.51	0.145	0.085	0.106	0.28	0.541	0.786
6	fischer-shofty_2013	0.07	-0.2	0.35	0.14	0.088	0.11	0.297	0.571	0.814
7	gibson_2014	-0.12	-1.13	0.89	0.515	0.053	0.054	0.067	0.09	0.121
8	gordon_2013	-0.15	-0.51	0.2	0.181	0.073	0.086	0.197	0.381	0.598
9	guastella_2010	0.59	0.07	1.12	0.268	0.06	0.066	0.116	0.201	0.321
10	guastella_2015	0.05	-0.54	0.64	0.301	0.058	0.063	0.102	0.169	0.264
11	jarskog_2017	-0.3	-0.83	0.23	0.27	0.06	0.066	0.115	0.199	0.316
12	woolley_2014	-0.01	-0.29	0.26	0.14	0.088	0.11	0.297	0.571	0.814

Note: Only effect sizes from 0.1 to 0.4 are shown here to preserve space. yi = effect size; lower = lower confidence interval bound; upper = upper confidence interval bound; power_es_observed = statistical power assuming that the observed summary effect size is the "true" effect size; power_es01 = statistical power assuming that 0.1 is the "true" effect size, and so forth.

using standard error data is recommended if both variance data types are available, as less data entry is required for the standard error approach compared to the confidence interval approach, which reduces the opportunity for data entry errors. As when using standard error data, the calculations in the **`mapower_ul()`** function assume that effects sizes and their corresponding 95% confidence intervals are normally distributed. Consequently, non-normally distributed effect sizes (e.g., Pearson's correlation coefficients, risk ratios, and odds ratios) will need to be transformed into normally distributed effect sizes (Borenstein et al., 2021; Harrer et al., 2021).

Visualizing study-level power across multiple meta-analyses

Comparing the median study-level statistical power across multiple analysis that use the same effect size metric is a useful way to evaluate the evidential value of research studies across fields or to compare different subfields. For example, the two previously generated Firepower plots, which both used Hedges' g as the effect size metric, can be combined into a single Firepower plot using the following script:

```
firepower(list(ooi_power_med_table, keech_power_med_table),
es = "Hedges' g")
```

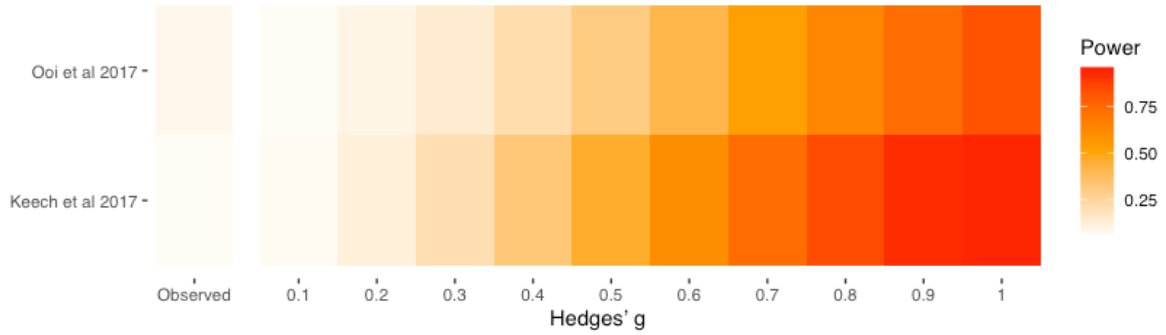


Fig. 6. Combining Firepower plots can facilitate the comparison of study-level statistical power for a range of effect sizes between meta-analyses. This plot reveals that the Keech and colleagues' (2018) meta-analysis contains studies that were designed to reliably detect a wider range of Hedges' g effect sizes, compared to the meta-analysis from Ooi and colleagues (2016).

This visualization demonstrates that the studies included in the Keech and colleagues' meta-analysis were designed to reliably detect a wider range of effect sizes than the studies in the Ooi and colleagues meta-analysis (**Fig. 6**). This approach can also be used when presenting results from multiple novel meta-analyses in the same article, as demonstrated by Boen and colleagues (2022).

Converting standardized mean differences to biserial correlation coefficients to facilitate effect size comparison

The previous section presented instructions for comparing study-level power across two or more meta-analyses that use the same effect size metric. However, in some situations a meta-analyst may want to synthesize data reported using different effect size metrics, which can make direct comparison difficult. A plausible scenario for the comparison of two different effect size metrics is the comparison of studies where researchers dichotomize one continuous variable into two groups—although this

practice in most circumstances has been the subject of critique (MacCallum et al., 2002)—with studies that evaluate the relationship between two continuous variables. In this situation, effect size conversion is required for comparing study-level statistical power for studies or meta-analyses that use standardized mean differences and correlation coefficients.

A common conversion approach transforms mean differences into a point-biserial correlation coefficient (e.g., Borenstein et al., 2021). However, this method has been shown to demonstrate bias (Jacobs & Viechtbauer, 2017). An alternative method that is largely free of bias is to transform mean difference data (means, standard deviations, and sample sizes per group) into a biserial correlation coefficient and its variance for comparison with Pearson's *r* and its variance (Jacobs & Viechtbauer, 2017). These correlation coefficients can then be reliably combined for meta-analysis and for the calculation of study-level statistical power using the *metameta* package (**Fig. 7**). This analysis process is demonstrated in Appendix B.

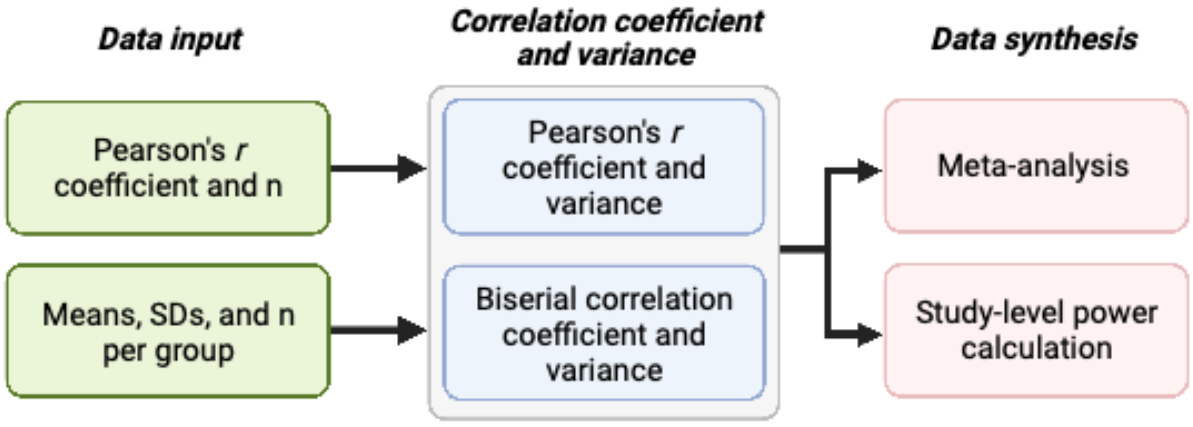


Fig. 7. The conversion of standardized mean differences to biserial correlation coefficients can facilitate data synthesis with Pearson's *r* coefficients for meta-analysis and the calculation of study-level statistical power.

Calculating study-level statistical power for new meta-analyses

It is relatively straightforward to integrate the calculation of study-level statistical power into the workflow of a novel meta-analysis using the popular *metafor* package (Viechtbauer, 2010). The **`escalc()`** function in *metafor* calculates effect sizes and their variances from information that is commonly reported (e.g., means).

To use this data in *metameta*, variance data needs to be converted into standard errors by calculating the square root of the effect size variances. Assuming that your datafile is named 'dat' and that the variances are in a column named 'vi', you can create a new column with standard errors (sei) using the following script: `dat$sei <- sqrt(dat$vi)`. This updated dataset with standard errors can now be used in the **`mapower_se()`** function.

Implications for summary effect size statistical power

One advantage of meta-analysis is that while individual included studies may not have sufficient statistical power to reliably detect a wide range of effect sizes, the synthesis of several of these studies into a summary effect size can increase statistical power. Indeed, a future meta-analysis has been proposed as potential justification for performing studies including small samples due to resource limitations, such as undergraduate student research projects or when collecting data from rare populations (Lakens, 2022; Quintana, 2021). However, under typical circumstances

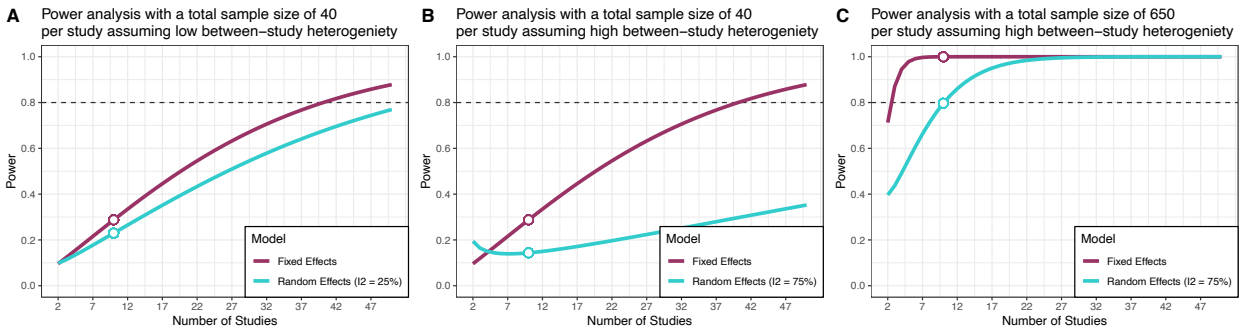


Fig. 8. Statistical power analysis for meta-analyses synthesizing between-participant experiments assuming a true effect size of 0.14 for three different scenarios. A random-effects meta-analysis with low heterogeneity ($I^2 = 25\%$), ten studies, and forty participants per study would achieve 24% statistical power (A). With higher heterogeneity ($I^2 = 75\%$), statistical power reduces to 14% (B). Assuming high heterogeneity ($I^2 = 75\%$), a sample size of 650 is required to achieve statistical power of 80% (C). The R script to reproduce these figures can be found on the article's OSF page: https://osf.io/dr64q/.

for the psychological sciences, meta-analysis is not a straightforward remedy for synthesizing underpowered studies, especially those that can only reliably detect large effect sizes, as increases in overall power via meta-analysis may be modest.

To illustrate this issue, consider the calculation of statistical power for a meta-analysis set of ten between-participant experimental studies with an average sample size of 40, low heterogeneity ($I^2 = 25\%$), and a true effect size of 0.14 (Quintana, 2020), which are parameters analogous to the examples described above. This analysis would suggest that such a research design would have 23% statistical power for a random-effects meta-analysis (**Fig. 8A**; for R code, see https://osf.io/dr64q/). At least 54 studies would be required to achieve 80% statistical power, holding these other parameters constant. Assuming high heterogeneity ($I^2 = 75\%$) with these original parameters, statistical power drops to 14% **(Fig. 8B**), which highlights the impact of study heterogeneity on statistical power. Statistical power for this meta-

analysis design with high heterogeneity ($I^2 = 75\%$) only reaches 80% power with a sample size of 650 (**Fig. 7C**).

Although I just demonstrated that a meta-analysis can theoretically rescue a body of underpowered studies when the effect size of small *if* there are a large enough number of studies, publication bias (Borenstien et al., 2009) and questionable research practices that tend to be associated with underpowered studies (Dwan et al., 2008) represent formidable problems unless the meta-analysis of small sample sizes and their constituent studies were pre-planned and there was a commitment to include study-level data regardless of statistical significance, which is currently rare in practice in the psychological sciences. Pre-planning included studies can also reduce between-study heterogeneity as study designs can be better harmonized (Halpern et al., 2002), which would increase meta-analysis statistical power. The Collaborative Replications and Education Project (CREP) is an example of pre-planned meta-analyses of psychology studies that would be statistically underpowered on their own (Wagge et al., 2019). This pre-planned meta-analysis approach is more common in medicine (e.g., Simes & The PPP and CTT Investigators, 1995), likely due to the similarity of this approach to multi-center clinical trials. For a retrospective meta-analysis, which is the norm in the psychological sciences, researchers are advised to use methods for the detection of publication bias and potential adjustment of the summary effect size due to publication bias (e.g., Robust Bayesian meta-analysis; Bartoš et al., 2022).

Summary

The *metameta* package can help evaluate the evidential value of studies included in a meta-analysis by calculating their statistical power. This package extends the existing sunset plot approach by calculating and visualizing statistical power assuming a range of effect sizes, rather than for a single effect size. This tool has been designed to use data that are commonly reported in meta-analysis forest plots—effect sizes and their variances. The increasing recognition of the importance of considering confidence in the body of evidence used in a meta-analysis is reflected in the inclusion of a checklist item on this topic in the recently updated PRISMA checklist (Page et al., 2021). By generating tables and visualizations, the *metameta* package is well suited to help authors and readers evaluate confidence in a body of evidence.

Statistical power is one of many approaches to evaluate the evidential value of a body of work and should not be used as a standalone proxy for study quality or for the overall quality of a meta-analysis. For example, the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) framework is a common tool for evaluating the quality of evidence in systematic reviews (Balshem et al., 2011), which considers five broad domains: risk of bias (e.g., study design and validity; Flake & Fried, 2020), inconsistency (i.e., heterogeneity; Higgins

& Thompson, 2002), indirectness (i.e., the representativeness of study samples; Ghai, 2021; Rad et al., 2018), imprecision (i.e., effect size variances), and publication bias (van Aert et al., 2019). As demonstrated above (**Fig. 8**), both inconsistency/heterogeneity and imprecision/variance can have a direct impact on the overall statistical power of a meta-analysis. Of these five domains, publication bias has historically been the most difficult to determine with confidence, as researchers need to make decisions about evidence that does not exist, at least in publicly (Guyatt et al., 2011). Various tools have more recently been developed for detecting and/or correcting for publication bias, such as Robust Bayesian meta-analysis (Bartoš et al., 2020), selection models (Maier et al., 2022; Vevea & Woods, 2005), p-curve (Simonsohn et al., 2014), and z-curve (Brunner & Schimmack, 2020). Another issue that can influence the evidential value of a body of work is the misreporting of statistical test results. Recently developed tools can evaluate the presence of reporting errors, such as GRIM (Brown & Heathers, 2017), SPRITE (Heathers et al., 2018), and statcheck (Nuijten & Polanin, 2020). These misreported statistical test results are quite common in psychology papers, with a 2016 study reporting that just under half of a sample of over 16,000 papers contained at least one statistical inconsistency, in which a *p*-value was not consistent with its test statistic and degrees of freedom (Nuijten et al., 2016). This is especially concerning for meta-analyses, as test statistics and *p*-values are sometimes used for calculating effect sizes and their variances (Lipsey & Wilson, 2001).

The main purpose of the *metameta* package is to determine the range of effect sizes that can be reliably detected for a body of studies. This tutorial used an 80% power criterion to determine reliability, however, other power levels can be used when justified. Indeed, the 80% power convention does not have a strong empirical basis, but rather, reflected the personal preference of Jacob Cohen (Cohen, 1988; Lakens, 2022). While a 20% Type II error rate (i.e., 80% statistical power) can be a good starting point judging the evidential value of a study, or body of studies, one should consider whether other Type II error rates for the research question at hand are more appropriate (Lakens, 2022; Maier & Lakens, 2022). For example, when working with rare study populations or when collecting observations is expensive, it can be difficult to design studies that can detect small effect sizes due to resource limitations as the use of large sample sizes is unrealistic in these cases. Alternatively, in other situations, error rates less than 20% are warranted or more realistic. A benefit of the *metameta* package is that by presenting power for a range of effects, the reader judge what they consider to be appropriate power based on the research question at hand and the available resources.

A key feature of the metameta package is that it is designed to use data that has been extracted from meta-analysis forest plots and tables, which is a much faster process than calculating effect size and variance data for each individual study. However, this approach assumes that meta-analysis data has been accurately extracted and calculated. For instance, standard errors may have been used instead

of standard deviations for meta-analysis calculations, which can influence reported effect sizes and variances (Kadlec et al., 2022). Using the free Zotero reference manager app (https://www.zotero.org/) can help mitigate this potential error as this app alerts users if they have imported a retracted meta-analysis article of if an article in their database is retracted after being imported. Users should also consider double-checking effect sizes that seem unrealistically large for the research field, which are often due to extraction or calculation errors (Kadlec et al., 2022).

The *metameta* package has been designed for the straightforward calculation of study-level statistical power and the median statistical power for a body of work when effect size and variance data is presented in published work or when researchers are reporting new meta-analysis. Conversely, this package is not designed to calculate statistical power for meta-analysis summary effect size estimates, heterogeneity tests, or moderator tests. However, resources to perform such tests are available elsewhere (Hedges & Pigott, 2004; Huedo-Medina et al., 2006; Valentine et al., 2010). The *metameta* package is also not designed to work with meta-analyses of nested data (e.g., when several effect sizes are extracted from the same study population), as this would bias the calculations for the median statistical power for a body of research. Another limitation of the package is that it assumes that the range of effect sizes of interest is greater than zero and less than a value of 2.5, which is the available range for analysis.

The overall goal of this tutorial is to provide an accessible guide for calculating and visualizing the study-level statistical power for meta-analyses for a range of effect sizes using the *metameta* R package. The companion video tutorial to this article provides additional guidance for readers who are not especially comfortable navigating R scripts. Alternatively, a point-and-click web app has also been provided for those without any programming experience.

Appendix A
Calculating power with effect sizes and sample sizes

The **`escalc()`** function from the *metafor* package (Viechtbauer, 2010) can calculate the variance of several effect size types when only sample size and effect size data is available. Below is a demonstration of how to calculate standard errors for Cohen's *d* and Pearson's *r*, which are among the most common effect sizes reported in the psychological sciences (for R code, see https://osf.io/dr64q/).

First, the calculation of the standard error of a Cohen's *d* value that was generated via a between-participants design if the sample sizes for each group have been reported will be demonstrated. If a Cohen's *d* value of 0.36 was calculated via the comparison of two independent groups (group 1 n = 93, group 2 n = 87), the following script will calculate the standard error values required for use in the *metameta* **`mapower_se()`** function:

```
R>   var <- escalc(
          measure="SMD",
          di=0.36,
          n1i=93,
          n2i=87
          )
       sqrt(var$vi)
```

It is important to reiterate that this function is not designed to calculate the variance of Cohen's *d* if this was generated via dependent groups (e.g., repeated measures). To calculate and standard errors from effect sizes and sample sizes for a within-participants design additionally requires the correlation coefficient for the

relationship of the variable of interest between groups, which is rarely reported in the psychological sciences.

Next, the calculation of Pearson's *r* standard error, if sample size information is available, will be exhibited. As Pearson's *r* can introduce bias when calculating variance in studies with small samples (Alexander et al., 1989), it is has been recommended to use Fisher's Z transformed values for the meta-analysis of correlational studies (Borenstein et al., 2021; Harrer et al., 2021; Quintana, 2015), which are normally distributed. The **`escalc()`** function in the *metafor* package can also transform Pearson's *r* values and their associated sample sizes into Fisher's Z scores and their variances. For example, the following script will calculate the standard error associated with a Pearson's *r* value of 0.22 that has been transformed into Fisher's Z, assuming a sample size of 112:

```
R>   var <- escalc(
          measure="ZCOR",
          ri=0.22,
          ni=112,
          )
     sqrt(var$vi)
```

This Fisher's Z values and their variances can then used in the **`mapower_se()`** function, as described above.

Appendix B

Transforming effect sizes

To illustrate the synthesis of mean difference data and correlation coefficient data, consider the five studies presented in **Table B1**. Three studies report Pearson's r and sample size (studies 1-3) and two studies report means, standard deviations, and sample sizes from two dichotomized and independent groups (studies 4-5). The first step is to transform the group comparison data from studies 4 and 5 into biserial correlation coefficients (r_b) and their variances (for R code, see https://osf.io/dr64q/), which can be reliably compared with Pearson's product–moment coefficients (r; Jacobs & Viechtbauer, 2017). A conventional random-effects meta-analysis can the performed on these coefficient effect sizes and their variances for data synthesis (**Fig. B1A**).

A variance-stablizing transformation is used for calculating the 95% confidence intervals, which is similar to Fisher's Z transformation and follows a normal distribution (Jacobs & Viechtbauer, 2017). These effect sizes and confidence

Table B1. Converting standardized mean differences to biserial correlation coefficients to facilitate effect size comparison

Study	r	n	m1	m2	sd1	sd2	n1	n2	yi	vi	measure	lowerCI	upperCI
Study 1	0.44	34							0.44	0.02	r	0.11	0.67
Study 2	0.75	112							0.75	0.01	r	0.65	0.82
Study 3	0.51	24							0.51	0.02	r	0.14	0.76
Study 4			9.46	7.91	3.73	2.74	78	81	0.29	0.01	r_b	0.1	0.46
Study 5			8.67	7.01	3.35	2.45	98	102	0.34	0.01	r_b	0.18	0.49

r = Pearson's r correlation coefficient, n = sample size for correlation, m1 = mean value for group 1, m2 = mean value for group 2, sd1 = standard deviation for group 1, sd2 = standard deviation for group 2, n1 = sample size for group 1, n2 = sample size for group 2, yi = effect size, vi = variance, measure = correlation coefficient measure used for effect size, lowerCI = lower bound for 95% confidence interval, upperCI = upper cound for 95% confidence interval, r_b = biserial correlation coefficient.

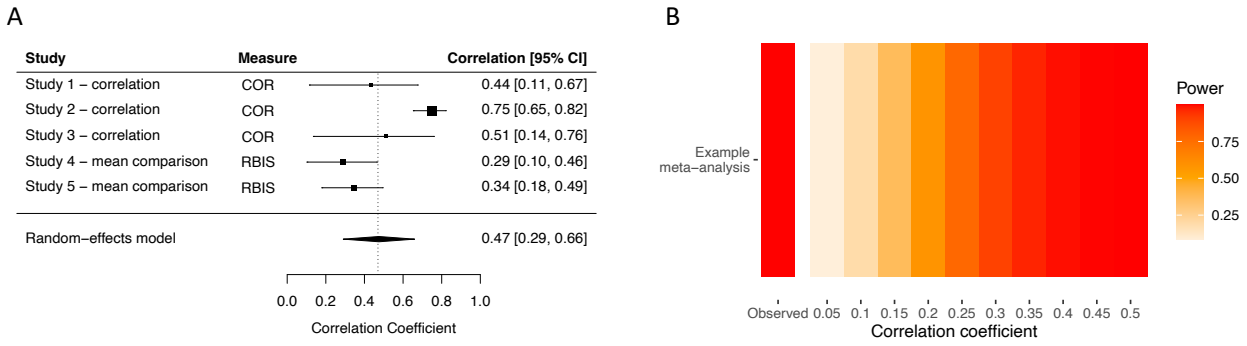


Fig. B1. The synthesis of mean comparison and correlation effect size data. Mean comparison data from studies 4 and 5 have been converted into biserial correlation coefficients (RBIS) and their variances. These effect sizes can be combined with the Pearson (product–moment) correlation coefficients (COR) from studies 1-3 for meta-analysis (A). Effect sizes and 95% confidence interval data (generated using variance-stablizing transformations) can be used to calculate the statistical median power for these studies for a range of effect sizes (B). The R script for reproducing these figures can be found on the article's OSF page: https://osf.io/dr64q/.

intervals can then be applied to the **`mapower_se()`** function for the calculation of study level statistical power (**Fig. B1B**; for R code, see https://osf.io/dr64q/). This analysis indicates that this body of studies could reliably detect (i.e., 80% power) correlation coefficients of approximately 0.3 and higher.

Authorship contributions

D. S. Quintana is the sole author of this manuscript and is responsible for its content. He developed the idea, wrote the article, wrote accompanying R scripts, created the accompanying video tutorial, and created all figures shown here.

Conflicts of Interest

The author declares that there were no conflicts of interest with respect to the authorship or the publication of this article.

Open practices

Open data: https://osf.io/dr64q/

Open materials: https://osf.io/dr64q/

Preregistration: Not applicable

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